

DSA Mastery: Practice & Concepts

Comprehensive MCQ Series for GATE, ISRO, BARC & DRDO Exams

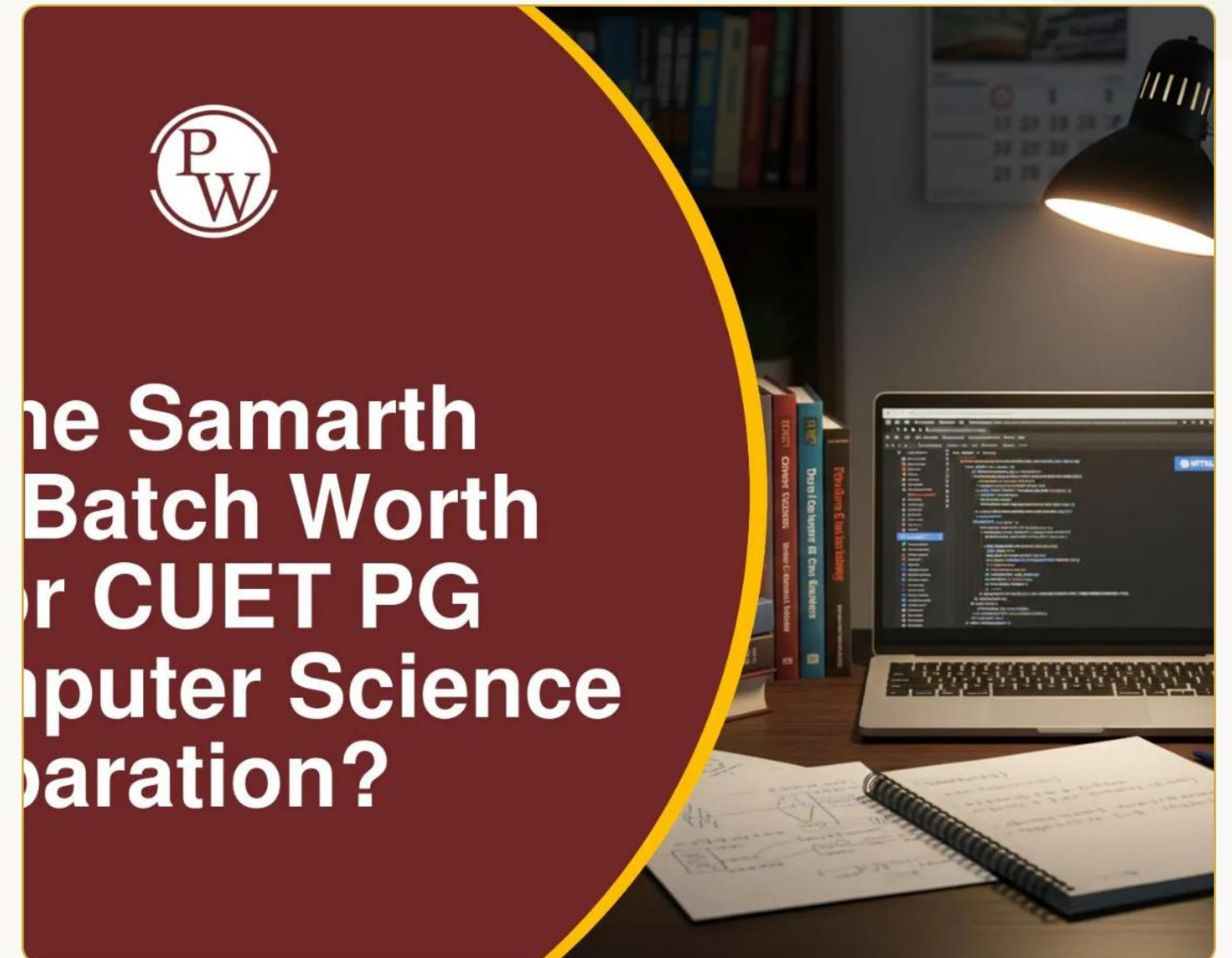
STACK • QUEUE • LINKED LIST

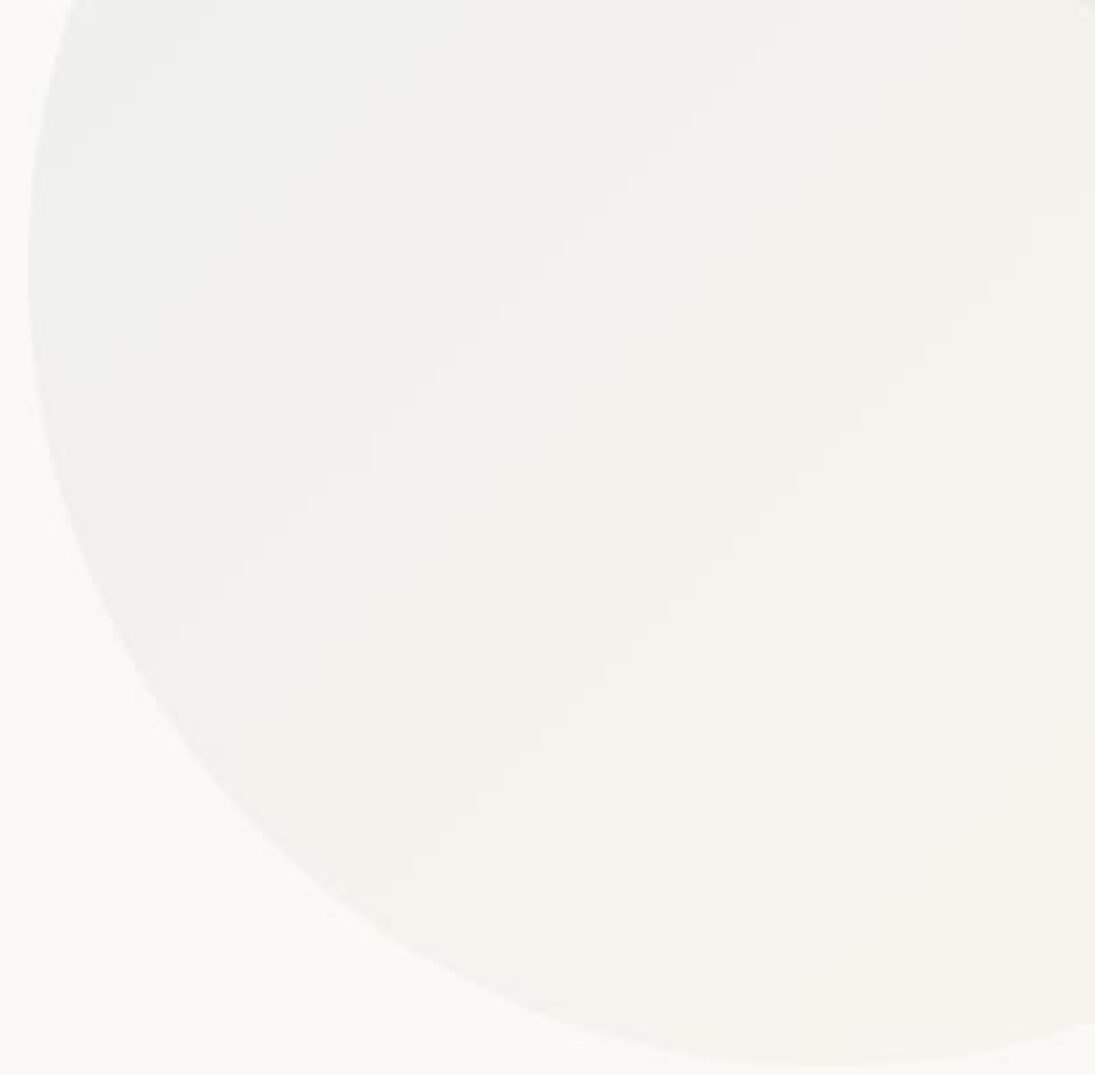
Exam-Centric Data Structures

This masterclass covers 45 crucial questions from previous year patterns of top-tier engineering exams. We focus on:

- **Stacks:** LIFO operations and applications.
- **Queues:** FIFO logic and implementation types.
- **Linked Lists:** Dynamic memory and node manipulation.

"Clear concepts are the foundation of problem-solving in competitive Computer Science."





PART I

STACK

LIFO Principle & Push/Pop Operations

Stack MCQs (Questions 1-8)

#	Question & Options	Ans	Explanation
1	Push(10), (20), (30) in MAX 5. Value of top? (A) 0 (B) 1 (C) 2 (D) 3	C	Starts from -1. Three pushes: -1 -> 0 -> 1 -> 2.
2	DS used for recursive function calls? (A) Queue (B) Stack (C) Linked List (D) Heap	B	Function recursion uses the Call Stack.
3	Postfix AB+C* equivalent? (A) A+(B*C) (B) (A+B)*C (C) A*(B+C) (D) (A*B)+C	B	AB+ yields (A+B), then *C yields (A+B)*C.
4	O(1) operation in stack array? (A) Search (B) Push (C) Traverse (D) Sort	B	Push is direct index assignment; O(1).
5	Overflow occurs when: (A) Empty (B) Full (C) Top=0 (D) Top=Negative	B	Overflow = attempting push to a full stack.
6	Infix to Postfix conversion uses? (A) Queue (B) Stack (C) Heap (D) Tree	B	Operators are pushed to a stack during conversion.
7	Linked List stack insertion complexity? (A) O(n) (B) O(log n) (C) O(1) (D) O(n log n)	C	Inserting at head is O(1) in Linked Lists.
8	Traversal implemented using Stack? (A) Level Order (B) Preorder (C) Breadth First (D) None	B	Preorder/DFS uses a stack for node storage.

Stack MCQs (Questions 9-15)

#	Question & Options	Ans	Explanation
9	Stack: 10, 20, 30 (top). After pop, top is? (A) 10 (B) 20 (C) 30 (D) NULL	B	<i>LIFO: Popping 30 reveals 20 as new top.</i>
10	Principle followed by Stack? (A) FIFO (B) LIFO (C) Random (D) Circular	B	<i>Last-In First-Out.</i>
11	Stacks required to evaluate postfix? (A) 2 (B) 1 (C) 3 (D) Op-dependent	B	<i>A single operand stack is sufficient.</i>
12	Balanced parentheses checking uses? (A) Queue (B) Stack (C) Graph (D) Heap	B	<i>Classic stack application for matching pairs.</i>
13	Pop operation deletes? (A) First (B) Middle (C) Last inserted (D) Random	C	<i>Pop removes the most recent element.</i>
14	Pointer storing top element? (A) head (B) rear (C) top (D) front	C	<i>'top' is the standard indexing pointer.</i>
15	Max elements in stack size 10? (A) 9 (B) 10 (C) 11 (D) Compiler-dependent	B	<i>A size-10 stack holds exactly 10 items.</i>

PART II

QUEUE

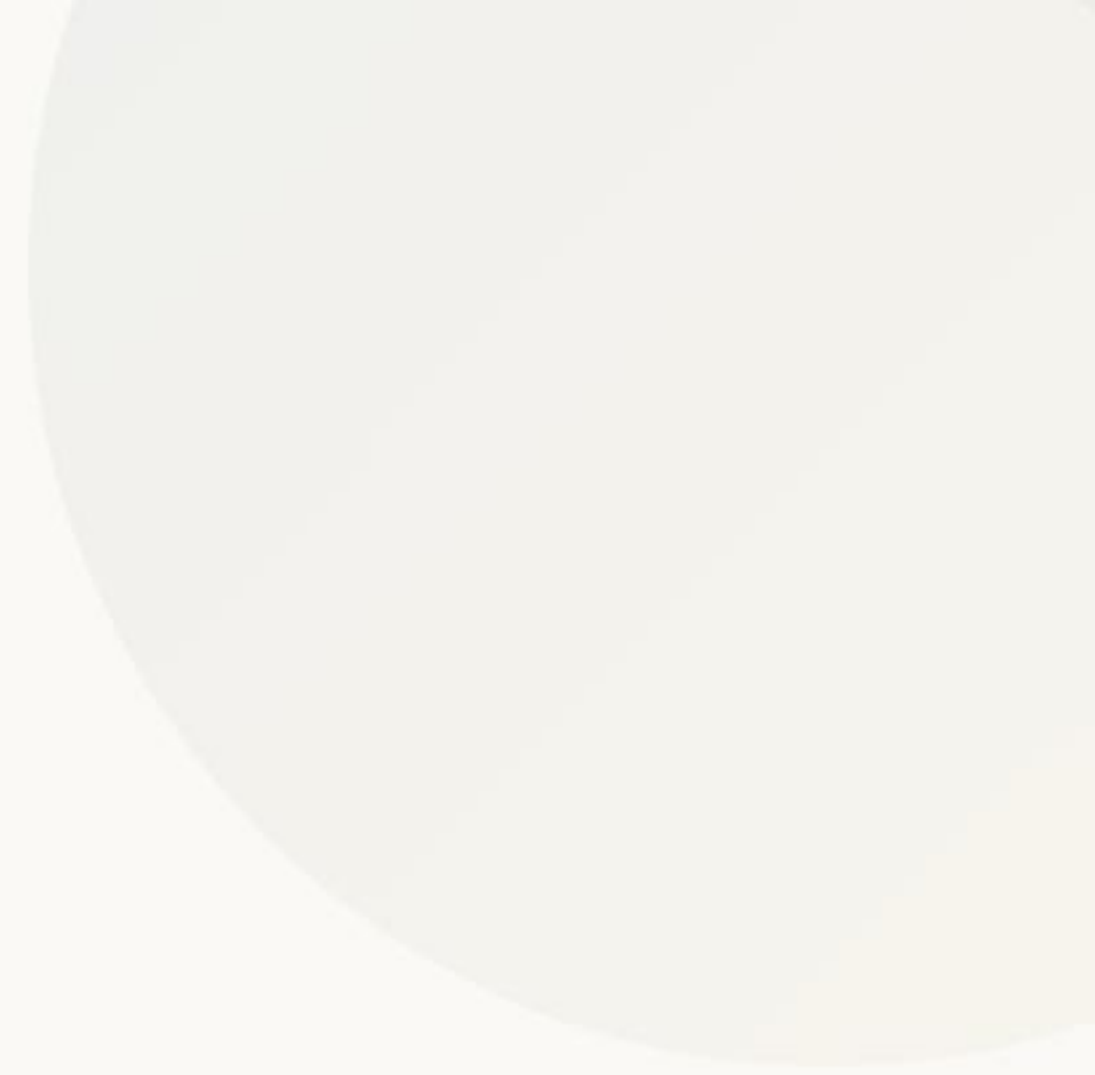
FIFO Principle & Linear vs Circular Queues

Queue MCQs (Questions 1-8)

#	Question & Options	Ans	Explanation
1	Queue follows? (A) LIFO (B) FIFO (C) Random (D) Priority	B	First-In First-Out principle.
2	Insertion in queue performed at? (A) Front (B) Rear (C) Middle (D) Anywhere	B	Elements join at the back (Rear).
3	Deletion from queue occurs at? (A) Rear (B) Front (C) Middle (D) Random	B	Elements leave from the start (Front).
4	Circular queue main use? (A) Memory (B) False overflow (C) Complexity (D) Speed	B	Allows reuse of empty spaces at the front.
5	Time complexity of enqueue? (A) $O(1)$ (B) $O(n)$ (C) $O(\log n)$ (D) $O(n^2)$	A	Constant time pointer update.
6	Application of Queue? (A) Function calls (B) CPU Scheduling (C) Evaluation (D) Recursion	B	Scheduling tasks/processes uses queues.
7	Pointer changing during enqueue? (A) Front (B) Rear (C) Top (D) Head	B	The Rear pointer advances on new entry.
8	Queue overflow occurs when? (A) Empty (B) Full (C) $\text{Front}=\text{Rear}$ (D) $\text{Front}=-1$	B	Overflow = attempting enqueue to full queue.

Queue MCQs (Questions 9–15)

#	Question & Options	Ans	Explanation
9	Priority Queue removes? (A) First (B) Highest priority (C) Last (D) Random	B	Removal based on weight/priority value.
10	Deque stands for? (A) Double Ended (B) Dynamic (C) Double (D) Directed	A	Allows I/O at both ends.
11	Traversal using Queue? (A) DFS (B) BFS (C) Inorder (D) Postorder	B	Breadth-First Search uses a queue.
12	Front pointer indicates? (A) Last (B) First (C) Middle (D) NULL	B	Front tracks the next element to exit.
13	Worst case enqueue complexity (linked queue)? (A) $O(1)$ (B) $O(\log n)$ (C) $O(n)$ (D) $O(n^2)$	A	With a tail pointer, it is $O(1)$.
14	Removes elements from both ends? (A) Priority (B) Circular (C) Deque (D) Linear	C	Double Ended Queue (Deque).
15	Circular Queue uses? (A) Modulo Arithmetic (B) Recursion (C) DP (D) Hashing	A	$i = (i+1) \% \text{SIZE}$ for wrap-around.



PART III

LINKED LIST

Dynamic Allocation & Node Traversal

Linked List MCQs (Questions 1-8)

#	Question & Options	Ans	Explanation
1	Dynamic memory allocation DS? (A) Array (B) Linked List (C) Stack (D) Queue	B	Nodes are allocated at runtime.
2	Singly linked list node contains? (A) Data (B) Pointer (C) Data+Pointer (D) Index	C	Structure holding value and next address.
3	Insertion at beginning complexity? (A) $O(1)$ (B) $O(\log n)$ (C) $O(n)$ (D) $O(n^2)$	A	Only head update required; $O(1)$.
4	Deleting first node requires? (A) Head update (B) Tail update (C) Sort (D) None	A	$head = head \rightarrow next$.
5	LL overcomes which array drawback? (A) Random (B) Fixed Size (C) Search (D) Sort	B	Linked lists grow/shrink dynamically.
6	Linked list traversal requires? (A) Index (B) Pointer (C) Queue (D) Stack	B	Pointers navigate from node to node.
7	Last node pointer contains? (A) Zero (B) NULL (C) Address of Head (D) Data	B	NULL signifies end of the list.
8	Circular LL last node points to? (A) NULL (B) First Node (C) Middle (D) Previous	B	The list forms a loop back to head.

Linked List MCQs (Questions 9-15)

#	Question & Options	Ans	Explanation
9	Searching in singly LL complexity? (A) $O(1)$ (B) $O(\log n)$ (C) $O(n)$ (D) $O(n^2)$	C	Linear search is mandatory; $O(n)$.
10	Doubly linked list contains? (A) 1 pointer (B) 2 pointers (C) 3 pointers (D) 0	B	Contains 'prev' and 'next' pointers.
11	Memory utilization? (A) Static (B) Dynamic (C) Fixed (D) Constant	B	Heaps are used for dynamic memory.
12	Difficult operation in Singly LL? (A) Fwd (B) Bwd (C) Insert Start (D) Delete Start	B	Singly LL has no 'prev' pointer.
13	Head pointer stores? (A) Last node (B) First node address (C) Middle (D) Data	B	Entry point of the linked list.
14	Loop detection algorithm? (A) Binary (B) Floyd's Cycle (C) Merge (D) BFS	B	Uses slow and fast pointers.
15	Insertion at end (no tail pointer)? (A) $O(1)$ (B) $O(\log n)$ (C) $O(n)$ (D) $O(n^2)$	C	Requires traversal to the last node.

Conclusion & Key Takeaways



LIFO vs FIFO

Understand the fundamental difference between Stacks (Last-In) and Queues (First-In) for algorithm selection.



Time Complexity

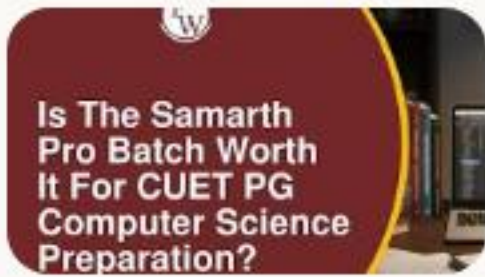
Mastering $O(1)$ vs $O(n)$ scenarios across arrays and linked implementations is critical for GATE/ISRO.



Applications

Recursion (Stack), BFS (Queue), and Dynamic Memory (Linked Lists) form the core of exam questions.

Image Sources



<https://static.pw.live/5eb393ee95fab7468a79d189/bd2db85f-d413-4d31-8a00-12b9cac22259.webp>

Source: www.pw.live